

Identification and Clinical Features of Scabies

However, it can be difficult to find burrows in early cases of scabies or after the patient has extensively excoriated the lesions. Identification of a burrow can be facilitated by rubbing a black felt-tip marker across an affected area. After the excess ink is wiped away with an alcohol pad, the burrow appears darker than the surrounding skin due to ink accumulation within it.

In infants younger than 2 years of age, the face and scalp can be involved, whereas in adults this is rare. Erythematous to violaceous pruritic nodules may appear in the axillae and flanks in children, and on the scrotum in men; these are thought to represent a hypersensitivity reaction to the mite. Such nodules can persist for weeks after successful eradication of the infection. Vesicles and bullae may develop, particularly on the palms and fingers.

Crusted Scabies

In crusted scabies, hyperkeratotic plaques develop diffusely on the palms and soles, with nail thickening and dystrophy. The rest of the skin typically appears diffusely xerotic. Pruritus is variable and may be absent. These patients are highly contagious, as their skin may harbor over a million mites, and they can serve as sources of outbreaks in healthcare settings.

Diagnosis

The definitive diagnosis of scabies is made by microscopic identification of the mite, eggs, or fecal pellets (scybala). This is achieved by placing a drop of mineral oil over a burrow and scraping longitudinally with a number 15 scalpel blade, taking care not to cause bleeding. The collected material is transferred to a glass slide and examined under low-power microscopy.

Other diagnostic methods include: - **Dermoscopy**, which allows in vivo visualization of the mite (see Chap. 5). - **Polymerase chain reaction (PCR)**

assays that detect mite DNA from skin scales. - **Skin biopsy**, which may show the mite transected in the stratum corneum.

Differential Diagnosis

(Refer to Box 208-1 for a list of conditions that may mimic scabies.)

Complications

Secondary bacterial infection (impetiginization) is common and usually responds well to topical or oral antibiotics, depending on the extent of involvement. However, more serious complications such as lymphangitis and septicemia can occur, particularly in crusted scabies. Post-streptococcal glomerulonephritis has been reported following *Streptococcus pyogenes* pyodermas associated with scabies.

Prognosis and Clinical Course

If left untreated, scabies can persist for many years. In immunocompromised individuals, the disease may progress to the crusted form, which requires aggressive treatment and infection control measures. With appropriate therapy, the prognosis is excellent, although pruritus and skin lesions may persist for several weeks due to ongoing hypersensitivity.